

SKILL WEEK-12

1 . To execute large pipelines, launch multiple jobs, measure system stability, monitor resource usage, detect failures, and analyze performance.

Aim : To implement a C program that creates multiple child processes, executes pipeline commands, monitors their execution, detects failures, and measures the overall execution time.

Program :

```
alekhya@Alekhyaireddy: ~/OS × + v
GNU nano 7.2
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <sys/time.h>

int main()
{
    int pipe1[2];
    int pipe2[2];

    pid_t p1, p2, p3;

    struct timeval start, end;

    gettimeofday(&start, NULL);

    printf("Large Pipeline Execution\n");
    printf("-----\n");

    /* Create pipes */
    if (pipe(pipe1) == -1 || pipe(pipe2) == -1)
    {
        perror("pipe");
        return 1;
    }

    /* First job: ls */
    p1 = fork();

    if (p1 < 0)
    {
        perror("fork");
        return 1;
    }

    if (p1 == 0)
    {
        dup2(pipe1[1], STDOUT_FILENO);

        close(pipe1[0]);
        close(pipe1[1]);
        close(pipe2[0]);
        close(pipe2[1]);

        execlp("ls", "ls", "-l", NULL);
    }
}
```

```
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    perror("exec ls");
    exit(1);
}

/* Second job: grep */
p2 = fork();
if (p2 < 0)
{
    perror("fork");
    return 1;
}

if (p2 == 0)
{
    dup2(pipe1[0], STDIN_FILENO);
    dup2(pipe2[1], STDOUT_FILENO);

    close(pipe1[0]);
    close(pipe1[1]);
    close(pipe2[0]);
    close(pipe2[1]);

    execlp("grep", "grep", ".c", NULL);

    perror("exec grep");
    exit(1);
}

/* Third job: sort */
p3 = fork();
if (p3 < 0)
{
    perror("fork");
    return 1;
}

if (p3 == 0)
{
    dup2(pipe2[0], STDIN_FILENO);

    close(pipe1[0]);
    close(pipe1[1]);
    close(pipe2[0]);
    close(pipe2[1]);
}
```

```
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/* Parent closes pipes */
close(pipe1[0]);
close(pipe1[1]);
close(pipe2[0]);
close(pipe2[1]);

printf("Three pipeline jobs launched.\n");
printf("Job 1: ls -l\n");
printf("Job 2: grep .c\n");
printf("Job 3: sort\n\n");

/* Wait for all jobs */
int status1, status2, status3;

waitpid(p1, &status1, 0);
waitpid(p2, &status2, 0);
waitpid(p3, &status3, 0);

gettimeofday(&end, NULL);

double elapsed =
    (end.tv_sec - start.tv_sec) +
    (end.tv_usec - start.tv_usec) / 1000000.0;

printf("\nPipeline execution completed.\n");

if (WIFEXITED(status1) && WEXITSTATUS(status1) == 0)
    printf("Job 1: SUCCESS\n");
else
    printf("Job 1: FAILED\n");

if (WIFEXITED(status2) && WEXITSTATUS(status2) == 0)
    printf("Job 2: SUCCESS\n");
else
    printf("Job 2: FAILED\n");

if (WIFEXITED(status3) && WEXITSTATUS(status3) == 0)
    printf("Job 3: SUCCESS\n");
else
    printf("Job 3: FAILED\n");

printf("\nTotal execution time: %.6f seconds\n", elapsed);

return 0;
}
```

Output :

```
alekhya@Alekhyareddy: ~/OS  X + v
alekhya@Alekhyareddy:~$ cd ~/OSSP
alekhya@Alekhyareddy:~/OSSP$ nano task_large_pipeline.c
alekhya@Alekhyareddy:~/OSSP$ gcc -Wall -Wextra task_large_pipeline.c -o task_large_pipeline
alekhya@Alekhyareddy:~/OSSP$ ./task_large_pipeline
Large Pipeline Execution
-----
Three pipeline jobs launched.
Job 1: ls -l
Job 2: grep .c
Job 3: sort

-rw-r--r-- 1 alekhya alekhya   77 Sep 22 14:17 expected_report.txt
-rw-r--r-- 1 alekhya alekhya  130 Sep 22 13:39 calculator.h
-rw-r--r-- 1 alekhya alekhya  169 Sep 22 13:41 calculator.c
-rw-r--r-- 1 alekhya alekhya  355 Sep 22 14:14 calculator_test.c
-rw-r--r-- 1 alekhya alekhya  420 Aug 13 11:45 main_loop.c
-rw-r--r-- 1 alekhya alekhya  443 Sep 22 14:07 task_valgrind_leak.c
-rw-r--r-- 1 alekhya alekhya  465 Sep 22 13:42 main_module.c
-rw-r--r-- 1 alekhya alekhya  662 Aug 17 13:53 week4_T2.c
-rw-r--r-- 1 alekhya alekhya  726 Sep 18 03:58 signal_handler.c
-rw-r--r-- 1 alekhya alekhya  769 Sep 22 14:03 task_valgrind.c
-rw-r--r-- 1 alekhya alekhya  940 Aug 17 06:02 task3_fork.c
-rw-r--r-- 1 alekhya alekhya  954 Aug 17 13:32 task4_states.c
-rw-r--r-- 1 alekhya alekhya 1018 Sep 18 03:50 client.c
-rw-r--r-- 1 alekhya alekhya 1275 Sep 18 03:48 server.c
-rw-r--r-- 1 alekhya alekhya 1376 Sep 18 03:32 ls_grep.c
-rw-r--r-- 1 alekhya alekhya 1464 Sep 22 13:44 calculator.o
-rw-r--r-- 1 alekhya alekhya 1541 Sep 22 13:50 task_error_handling.c
-rw-r--r-- 1 alekhya alekhya 1550 Sep 21 04:56 task_signal_actions.c
-rw-r--r-- 1 alekhya alekhya 1617 Sep 21 05:54 task_signal_behavior.c
-rw-r--r-- 1 alekhya alekhya 1661 Aug  7 08:09 task3_input.c
-rw-r--r-- 1 alekhya alekhya 1677 Sep 18 03:26 producer_consumer.c
-rw-r--r-- 1 alekhya alekhya 1705 Sep 18 04:13 task2_single_quotes.c
-rw-r--r-- 1 alekhya alekhya 1851 Aug 17 13:42 task1_wait.c
-rw-r--r-- 1 alekhya alekhya 1851 Aug 17 13:46 task1_wait.c.save
-rw-r--r-- 1 alekhya alekhya 1959 Sep 21 05:15 task_sigint_forward.c
-rw-r--r-- 1 alekhya alekhya 2092 Aug 17 05:03 task5_process.c
-rw-r--r-- 1 alekhya alekhya 2131 Sep 21 05:09 task_sigint_forward.c.save
-rw-r--r-- 1 alekhya alekhya 2234 Sep 21 06:01 task_resume_job.c
-rw-r--r-- 1 alekhya alekhya 2297 Aug 17 04:59 task4_escape.c
-rw-r--r-- 1 alekhya alekhya 2380 Aug 13 11:56 task2_quotes.c
-rw-r--r-- 1 alekhya alekhya 2450 Sep 21 05:39 task_process_groups.c
-rw-r--r-- 1 alekhya alekhya 2602 Sep 21 05:29 task_sigstcp.c
-rw-r--r-- 1 alekhya alekhya 2612 Aug  7 08:30 task5_memory.c
-rw-r--r-- 1 alekhya alekhya 2705 Sep 22 14:27 task_large_pipeline.c
-rw-r--r-- 1 alekhya alekhya 3056 Sep 18 04:11 week_5_double_quotes.c
-rw-r--r-- 1 alekhya alekhya 3056 Sep 18 04:38 task3_double_quotes.c
-rw-r--r-- 1 alekhya alekhya 3976 Aug  7 08:19 task4_history.c
-rwxr-xr-x 1 alekhya alekhya  1187 Sep 22 14:16 test_script.sh
-rwxr-xr-x 1 alekhya alekhya 16064 Sep 22 14:14 calculator_test
-rwxr-xr-x 1 alekhya alekhya 16368 Sep 21 04:56 task_signal_actions
-rwxr-xr-x 1 alekhya alekhya 16392 Sep 18 03:51 client
-rwxr-xr-x 1 alekhya alekhya 16488 Aug 17 04:59 task4_escape
-rwxr-xr-x 1 alekhya alekhya 16488 Sep 18 03:26 producer_consumer
-rwxr-xr-x 1 alekhya alekhya 16648 Sep 21 05:39 task_process_groups
-rwxr-xr-x 1 alekhya alekhya 16696 Aug 17 05:03 task5_process
drwxr-xr-x 2 alekhya alekhya  4096 Aug  1 09:07 include
```



alekhya@Alekhyaireddy: ~/OS



```
drwxr-xr-x 2 alekhya alekhya 4096 Aug 1 09:07 include
drwxr-xr-x 2 alekhya alekhya 4096 Aug 1 09:08 docs
drwxr-xr-x 2 alekhya alekhya 4096 Aug 1 09:08 scripts
drwxr-xr-x 2 alekhya alekhya 4096 Aug 4 05:32 src
prw-r--r-- 1 alekhya alekhya 0 Sep 18 03:52 client_to_server
prw-r--r-- 1 alekhya alekhya 0 Sep 18 03:52 server_to_client
```

Pipeline execution completed.

Job 1: SUCCESS

Job 2: SUCCESS

Job 3: SUCCESS

Total execution time: 0.061923 seconds

alekhya@Alekhyaireddy:~/OSSP\$ ps -ef | grep task_large_pipeline

alekhya 716 677 0 14:27 pts/0 00:00:00 grep --color=auto task_large_pipeline

alekhya@Alekhyaireddy:~/OSSP\$ top

top - 14:27:46 up 48 min, 1 user, load average: 0.06, 0.04, 0.00

Tasks: 22 total, 1 running, 21 sleeping, 0 stopped, 0 zombie

%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 99.8 id, 0.2 wa, 0.0 hi, 0.0 si, 0.0 st

MiB Mem : 15743.6 total, 15097.3 free, 590.8 used, 251.4 buff/cache

MiB Swap: 4096.0 total, 4096.0 free, 0.0 used. 15152.8 avail Mem

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	21736	13228	9768	S	0.0	0.1	0:00.38	systemd
2	root	20	0	3180	2208	2072	S	0.0	0.0	0:00.00	init-systemd(Ub
6	root	20	0	3196	2132	2008	S	0.0	0.0	0:00.00	init
49	root	19	-1	50440	16248	15160	S	0.0	0.1	0:00.17	systemd-journal
101	root	20	0	25624	6752	5136	S	0.0	0.0	0:00.57	systemd-udev
114	systemd+	20	0	21344	13156	11032	S	0.0	0.1	0:00.05	systemd-resolve
115	systemd+	20	0	91036	7964	6996	S	0.0	0.0	0:00.05	systemd-timesyn
164	root	20	0	4244	2688	2432	S	0.0	0.0	0:00.00	cron
168	message+	20	0	9648	5412	4700	S	0.0	0.0	0:00.03	dbus-daemon
176	root	20	0	17976	8792	7748	S	0.0	0.1	0:00.03	systemd-logind
195	syslog	20	0	222516	5636	4592	S	0.0	0.0	0:00.04	rsyslogd
198	root	20	0	3124	1960	1824	S	0.0	0.0	0:00.00	agetty
199	root	20	0	107036	23196	13744	S	0.0	0.1	0:00.07	unattended-upgr
288	root	20	0	6696	4692	3916	S	0.0	0.0	0:00.01	login
331	alekhya	20	0	20320	11512	9392	S	0.0	0.1	0:00.04	systemd
332	alekhya	20	0	21168	3596	1868	S	0.0	0.0	0:00.00	(sd-pam)
351	alekhya	20	0	6080	5396	3708	S	0.0	0.0	0:00.01	bash
633	root	20	0	1756628	13704	11448	S	0.0	0.1	0:00.12	wsl-pro-service
673	root	20	0	3184	1120	980	S	0.0	0.0	0:00.00	SessionLeader
675	root	20	0	3200	1256	1108	S	0.0	0.0	0:00.28	Relay(677)
677	alekhya	20	0	6208	5408	3652	S	0.0	0.0	0:00.02	bash
717	alekhya	20	0	9392	5832	3604	R	0.0	0.0	0:00.01	top

```
[1]+ Stopped top
alekhya@AlekhyaReddy:~/OSSP$ ps -eo pid,ppid,%cpu,%mem,state,cmd
PID    PPID  %CPU  %MEM  S CMD
1      0    0.0   0.0   S /sbin/init
2      1    0.0   0.0   S /init
6      2    0.0   0.0   S plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
49     1    0.0   0.1   S /usr/lib/systemd/systemd-journald
101    1    0.0   0.0   S /usr/lib/systemd/systemd-udev
114    1    0.0   0.0   S /usr/lib/systemd/systemd-resolved
115    1    0.0   0.0   S /usr/lib/systemd/systemd-timesyncd
164    1    0.0   0.0   S /usr/sbin/cron -f -P
168    1    0.0   0.0   S @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
176    1    0.0   0.0   S /usr/lib/systemd/systemd-logind
195    1    0.0   0.0   S /usr/sbin/rsyslogd -n -iNONE
198    1    0.0   0.0   S /sbin/agetty -o -p -- \u --noclear - linux
199    1    0.0   0.1   S /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
288    2    0.0   0.0   S /bin/login -f
331    1    0.0   0.0   S /usr/lib/systemd/systemd --user
332    331  0.0   0.0   S (sd-pam)
351    288  0.0   0.0   S -bash
633    1    0.0   0.0   S /usr/libexec/wsl-pro-service
673    2    0.0   0.0   S /init
675    673  0.1   0.0   S /init
677    675  0.0   0.0   S -bash
717    677  0.0   0.0   T top
718    101  0.0   0.0   S (udev-worker)
719    101  0.0   0.0   S (udev-worker)
720    677  0.0   0.0   R ps -eo pid,ppid,%cpu,%mem,state,cmd
alekhya@AlekhyaReddy:~/OSSP$ free -h
              total        used        free      shared  buff/cache   available
Mem:           15Gi         591Mi        14Gi         3.5Mi        251Mi        14Gi
Swap:           4.0Gi           0B         4.0Gi
alekhya@AlekhyaReddy:~/OSSP$ uptime
 14:28:07 up 49 min,  1 user,  load average: 0.04, 0.03, 0.00
alekhya@AlekhyaReddy:~/OSSP$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            7.7G   0 7.7G   0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none            7.7G  4.0K 7.7G   1% /mnt/wsl
drivers         932G  184G 749G  20% /usr/lib/wsl/drivers
/dev/sdd        1007G   3.0G 953G   1% /
none            7.7G   44K 7.7G   1% /mnt/wslg
none            7.7G   0 7.7G   0% /usr/lib/wsl/lib
rootfs          7.7G   2.8M 7.7G   1% /init
none            7.7G  508K 7.7G   1% /run
none            7.7G   0 7.7G   0% /run/lock
none            7.7G   0 7.7G   0% /run/shm
none            7.7G   80K 7.7G   1% /mnt/wslg/versions.txt
none            7.7G   80K 7.7G   1% /mnt/wslg/doc
C:\             932G  184G 749G  20% /mnt/c
tmpfs           1.6G   20K 1.6G   1% /run/user/1000
alekhya@AlekhyaReddy:~/OSSP$ ./task_large_pipeline
Large Pipeline Execution
-----
Three pipeline jobs launched.
```



alekhya@Alekhyareddy: ~/OS



```
-rw-r--r-- 1 alekhya alekhya 77 Sep 22 14:17 expected_report.txt
-rw-r--r-- 1 alekhya alekhya 130 Sep 22 13:39 calculator.h
-rw-r--r-- 1 alekhya alekhya 169 Sep 22 13:41 calculator.c
-rw-r--r-- 1 alekhya alekhya 355 Sep 22 14:14 calculator_test.c
-rw-r--r-- 1 alekhya alekhya 420 Aug 13 11:45 main_loop.c
-rw-r--r-- 1 alekhya alekhya 443 Sep 22 14:07 task_valgrind_leak.c
-rw-r--r-- 1 alekhya alekhya 465 Sep 22 13:42 main_module.c
-rw-r--r-- 1 alekhya alekhya 662 Aug 17 13:53 week4_T2.c
-rw-r--r-- 1 alekhya alekhya 726 Sep 18 03:58 signal_handler.c
-rw-r--r-- 1 alekhya alekhya 769 Sep 22 14:03 task_valgrind.c
-rw-r--r-- 1 alekhya alekhya 940 Aug 17 06:02 task3_fork.c
-rw-r--r-- 1 alekhya alekhya 954 Aug 17 13:32 task4_states.c
-rw-r--r-- 1 alekhya alekhya 1018 Sep 18 03:50 client.c
-rw-r--r-- 1 alekhya alekhya 1275 Sep 18 03:48 server.c
-rw-r--r-- 1 alekhya alekhya 1376 Sep 18 03:32 ls_grep.c
-rw-r--r-- 1 alekhya alekhya 1464 Sep 22 13:44 calculator.o
-rw-r--r-- 1 alekhya alekhya 1541 Sep 22 13:50 task_error_handling.c
-rw-r--r-- 1 alekhya alekhya 1550 Sep 21 04:56 task_signal_actions.c
-rw-r--r-- 1 alekhya alekhya 1617 Sep 21 05:54 task_signal_behavior.c
-rw-r--r-- 1 alekhya alekhya 1661 Aug 7 08:09 task3_input.c
-rw-r--r-- 1 alekhya alekhya 1677 Sep 18 03:26 producer_consumer.c
-rw-r--r-- 1 alekhya alekhya 1705 Sep 18 04:13 task2_single_quotes.c
-rw-r--r-- 1 alekhya alekhya 1851 Aug 17 13:42 task1_wait.c
-rw-r--r-- 1 alekhya alekhya 1851 Aug 17 13:46 task1_wait.c.save
-rw-r--r-- 1 alekhya alekhya 1959 Sep 21 05:15 task_sigint_forward.c
-rw-r--r-- 1 alekhya alekhya 2092 Aug 17 05:03 task5_process.c
-rw-r--r-- 1 alekhya alekhya 2131 Sep 21 05:09 task_sigint_forward.c.save
-rw-r--r-- 1 alekhya alekhya 2234 Sep 21 06:01 task_resume_job.c
-rw-r--r-- 1 alekhya alekhya 2297 Aug 17 04:59 task4_escape.c
-rw-r--r-- 1 alekhya alekhya 2380 Aug 13 11:56 task2_quotes.c
-rw-r--r-- 1 alekhya alekhya 2450 Sep 21 05:39 task_process_groups.c
-rw-r--r-- 1 alekhya alekhya 2602 Sep 21 05:29 task_sigsttp.c
-rw-r--r-- 1 alekhya alekhya 2612 Aug 7 08:30 task5_memory.c
-rw-r--r-- 1 alekhya alekhya 2705 Sep 22 14:27 task_large_pipeline.c
-rw-r--r-- 1 alekhya alekhya 3056 Sep 18 04:11 week_5_double_quotes.c
-rw-r--r-- 1 alekhya alekhya 3056 Sep 18 04:38 task3_double_quotes.c
-rw-r--r-- 1 alekhya alekhya 3976 Aug 7 08:19 task4_history.c
-rwxr-xr-x 1 alekhya alekhya 1187 Sep 22 14:16 test_script.sh
-rwxr-xr-x 1 alekhya alekhya 16064 Sep 22 14:14 calculator_test
-rwxr-xr-x 1 alekhya alekhya 16368 Sep 21 04:56 task_signal_actions
-rwxr-xr-x 1 alekhya alekhya 16392 Sep 18 03:51 client
-rwxr-xr-x 1 alekhya alekhya 16488 Aug 17 04:59 task4_escape
-rwxr-xr-x 1 alekhya alekhya 16488 Sep 18 03:26 producer_consumer
-rwxr-xr-x 1 alekhya alekhya 16648 Sep 21 05:39 task_process_groups
-rwxr-xr-x 1 alekhya alekhya 16696 Aug 17 05:03 task5_process
drwxr-xr-x 2 alekhya alekhya 4096 Aug 1 09:07 include
drwxr-xr-x 2 alekhya alekhya 4096 Aug 1 09:08 docs
drwxr-xr-x 2 alekhya alekhya 4096 Aug 1 09:08 scripts
drwxr-xr-x 2 alekhya alekhya 4096 Aug 4 05:32 src
prw-r--r-- 1 alekhya alekhya 0 Sep 18 03:52 client_to_server
prw-r--r-- 1 alekhya alekhya 0 Sep 18 03:52 server_to_client
```

Pipeline execution completed.

Job 1: SUCCESS

Job 2: SUCCESS

Job 3: SUCCESS

Total execution time: 0.003449 seconds

real 0m0.006s

user 0m0.002s

sys 0m0.004s

alekhya@Alekhyareddy:~/OSSP\$ |

2 . To document the system architecture, create flow diagrams, explain execution paths, describe system calls, record design decisions, and review the project documentation.

Aim : To document the architecture and execution flow of the system, identify the system calls used, record important design decisions, and maintain clear and organized project documentation.

Program :

```
alekhya@Alekhyareddy: ~/OS × + v
GNU nano 7.2
SYSTEM ARCHITECTURE

1. User Interface
  |
  v
2. Command Parser
  |
  v
3. Command Executor
  |
  v
4. Process Management
  |
  v
5. Linux Kernel
  |
  v
6. Child Process / Program

Main Components:
- Parser: Reads and analyzes commands.
- Executor: Executes requested programs.
- Process Manager: Creates and manages processes.
- Signal Handler: Handles process signals.
- Linux Kernel: Provides system-call services.
```



alekhya@Alekhyareddy: ~/OS



GNU nano 7.2

EXECUTION FLOW

Start

|

v

Read Command

|

v

Validate Command

|

v

Parse Arguments

|

v

Create Child Process using fork()

|

v

Execute Program using exec()

|

v

Parent waits using wait()/waitpid()

|

v

Check Exit Status

|

v

Display Result

|

v

End

```
alekhya@Alekhyareddy: ~/OS × + ▾
GNU nano 7.2
SYSTEM CALLS

fork()
- Creates a new child process.

exec()
- Replaces the child process with a new program.

wait()
- Waits for a child process to finish.

waitpid()|
- Waits for a specific child process.

pipe()
- Creates communication between processes.

dup2()
- Redirects file descriptors.

kill()
- Sends a signal to a process.

signal()
- Registers a signal handler.

sigaction()
- Configures signal handling.

open()
- Opens a file.

close()
- Closes a file descriptor.
```

```
alekhya@Alekhyareddy: ~/OS × + ▾
GNU nano 7.2 docs/
DESIGN DECISIONS

1. Use modular C files for better organization.
2. Use fork() to create child processes.
3. Use exec() to execute external programs.
4. Use waitpid() to synchronize parent and child processes.
5. Use pipes for inter-process communication.
6. Use signal handling for process control.
7. Use separate documentation files for architecture,
   execution flow, system calls, and design decisions.
8. Use compiler warnings during compilation to improve reliability.
```

Output :

```
alekhya@Alekhyareddy: ~/OS × + v
alekhya@Alekhyareddy:~$ cd ~/OSSP
alekhya@Alekhyareddy:~/OSSP$ mkdir -p docs
alekhya@Alekhyareddy:~/OSSP$ nano docs/architecture.txt
alekhya@Alekhyareddy:~/OSSP$ nano docs/execution_flow.txt
alekhya@Alekhyareddy:~/OSSP$ nano docs/syscalls.txt
alekhya@Alekhyareddy:~/OSSP$ nano docs/design_decisions.txt
alekhya@Alekhyareddy:~/OSSP$ cat docs/architecture.txt
SYSTEM ARCHITECTURE

1. User Interface
   |
   v
2. Command Parser
   |
   v
3. Command Executor
   |
   v
4. Process Management
   |
   v
5. Linux Kernel
   |
   v
6. Child Process / Program

Main Components:
- Parser: Reads and analyzes commands.
- Executor: Executes requested programs.
- Process Manager: Creates and manages processes.
- Signal Handler: Handles process signals.
- Linux Kernel: Provides system-call services.
```

```
alekhya@Alekhyareddy:~/OSSP$ cat docs/execution_flow.txt
EXECUTION FLOW

Start
 |
 v
Read Command
 |
 v
Validate Command
 |
 v
Parse Arguments
 |
 v
Create Child Process using fork()
 |
 v
Execute Program using exec()
 |
 v
Parent waits using wait()/waitpid()
 |
 v
Check Exit Status
 |
 v
Display Result
 |
 v
End
```

```
alekhya@Alekhyareddy:~/OSSP$ cat docs/syscalls.txt  
SYSTEM CALLS
```

```
fork()  
- Creates a new child process.  
  
exec()  
- Replaces the child process with a new program.  
  
wait()  
- Waits for a child process to finish.  
  
waitpid()  
- Waits for a specific child process.  
  
pipe()  
- Creates communication between processes.  
  
dup2()  
- Redirects file descriptors.  
  
kill()  
- Sends a signal to a process.  
  
signal()  
- Registers a signal handler.  
  
sigaction()  
- Configures signal handling.  
  
open()  
- Opens a file.  
  
close()  
- Closes a file descriptor.
```

```
alekhya@Alekhyareddy:~/OSSP$ cat docs/design_decisions.txt  
DESIGN DECISIONS
```

1. Use modular C files for better organization.
2. Use fork() to create child processes.
3. Use exec() to execute external programs.
4. Use waitpid() to synchronize parent and child processes.
5. Use pipes for inter-process communication.
6. Use signal handling for process control.
7. Use separate documentation files for architecture, execution flow, system calls, and design decisions.
8. Use compiler warnings during compilation to improve reliability.

```
alekhya@Alekhyareddy:~/OSSP$ tree docs
docs
├── architecture.txt
├── design.md
├── design_decisions.txt
├── execution_flow.txt
├── report.pdf
└── syscalls.txt

1 directory, 6 files
alekhya@Alekhyareddy:~/OSSP$ |
```

3 . To organize source code, improve build scripts, add a README file, verify repository structure, tag releases, and validate project deliverables.

Aim : To maintain a well-organized project repository with a reliable build process, clear documentation, proper version tags, and verified project deliverables.

Program :

```
alekhya@Alekhyareddy: ~/OS  ×  +  ▾
GNU nano 7.2
# OSSP Project

## Description
This project demonstrates Operating System and Systems Programming concepts using C.

## Project Structure
- src/          - Source code
- include/     - Header files
- obj/         - Object files
- bin/         - Executable files
- docs/        - Documentation
- tests/       - Test programs
- scripts/     - Shell scripts

## Build
Use:
make

## Run
Use:
./bin/my_shell

## Clean
Use:
make clean

## Testing
Test programs are stored in the tests directory.

## Version
v1.0.0
```

```
alekhya@Alekhyareddy: ~/OS  ×  +  ∨
GNU nano 7.2
CC = gcc
CFLAGS = -Wall -Wextra -Iinclude

TARGET = bin/my_shell

SOURCES = src/main.c \
          src/shell.c \
          src/parser.c \
          src/executor.c \
          src/builtin.c

OBJECTS = $(SOURCES:src/%.c=obj/%.o)

all: $(TARGET)

$(TARGET): $(OBJECTS)
    mkdir -p bin
    $(CC) $(OBJECTS) -o $(TARGET)

obj/%.o: src/%.c
    mkdir -p obj
    $(CC) $(CFLAGS) -c $< -o $@

clean:
    rm -rf obj/*.o bin/my_shell

test: $(TARGET)
    @echo "Running project tests..."
    @echo "Tests completed."

.PHONY: all clean test
```

Output :



alekhya@Alekhyareddy: ~/OS



```
alekhya@Alekhyareddy:~$ cd ~/OSSP
alekhya@Alekhyareddy:~/OSSP$ git status
On branch master
```

No commits yet

Untracked files:

(use "git add <file>..." to include in what will be committed)

```
.gitignore
LICENSE
Makefile
README.md
assets/
bin/
calculator.c
calculator.h
calculator.o
calculator_test
calculator_test.c
client
client.c
docs/
error.log
expected_report.txt
include/
ls_grep
ls_grep.c
main_loop
main_loop.c
main_module.c
main_module.o
module_test
producer_consumer
producer_consumer.c
scripts/
server
server.c
signal_handler
signal_handler.c
src/
task1_wait
task1_wait.c
task1_wait.c.save
task2_quotes
task2_quotes.c
task2_single_quotes
task2_single_quotes.c
```



alekhya@Alekhyareddy: ~/OS



```
task3_double_quotes
task3_double_quotes.c
task3_fork
task3_fork.c
task3_input
task3_input.c
task4_escape
task4_escape.c
task4_history
task4_history.c
task4_states
task4_states.c
task5_memory
task5_memory.c
task5_process
task5_process.c
task_error_handling
task_error_handling.c
task_large_pipeline
task_large_pipeline.c
task_process_groups
task_process_groups.c
task_resume_job
task_resume_job.c
task_sigint_forward
task_sigint_forward.c
task_sigint_forward.c.save
task_signal_actions
task_signal_actions.c
task_signal_behavior
task_signal_behavior.c
task_sigtstp
task_sigtstp.c
task_valgrind
task_valgrind.c
task_valgrind_leak
task_valgrind_leak.c
test_report.txt
test_script.sh
tests/
valgrind_report.txt
week4_T2
week4_T2.c
week_5_double_quotes.c
week_5double_quotes
```

nothing added to commit but untracked files present (use "git add" to track)

alekhya@Alekhyareddy: ~/OS

+

▼

nothing added to commit but untracked files present (use "git add" to track)

alekhya@Alekhyareddy:~/OSSP\$ mkdir -p src include obj bin docs tests scripts

alekhya@Alekhyareddy:~/OSSP\$ tree

```
.
├── LICENSE
├── Makefile
├── README.md
├── assets
│   └── architecture.png
├── bin
│   ├── fork_exec
│   └── my_shell
├── calculator.c
├── calculator.h
├── calculator.o
├── calculator_test
├── calculator_test.c
├── client
├── client.c
├── client_to_server
├── docs
│   ├── architecture.txt
│   ├── design.md
│   ├── design_decisions.txt
│   ├── execution_flow.txt
│   ├── report.pdf
│   └── syscalls.txt
├── error.log
├── expected_report.txt
├── include
│   ├── builtin.h
│   ├── executor.h
│   ├── parser.h
│   └── shell.h
├── ls_grep
├── ls_grep.c
├── main_loop
├── main_loop.c
├── main_module.c
├── main_module.o
├── module_test
├── obj
├── producer_consumer
├── producer_consumer.c
├── scripts
│   └── run.sh
└── server
```



alekhya@Alekhyareddy: ~/OS



```
server
server.c
server_to_client
signal_handler
signal_handler.c
src
├── builtin.c
├── executor.c
├── fork_exec.c
├── main.c
├── parser.c
└── shell.c
task1_wait
task1_wait.c
task1_wait.c.save
task2_quotes
task2_quotes.c
task2_single_quotes
task2_single_quotes.c
task3_double_quotes
task3_double_quotes.c
task3_fork
task3_fork.c
task3_input
task3_input.c
task4_escape
task4_escape.c
task4_history
task4_history.c
task4_states
task4_states.c
task5_memory
task5_memory.c
task5_process
task5_process.c
task_error_handling
task_error_handling.c
task_large_pipeline
task_large_pipeline.c
task_process_groups
task_process_groups.c
task_resume_job
task_resume_job.c
task_sigint_forward
task_sigint_forward.c
task_sigint_forward.c.save
task_signal_actions
```

```

alekhya@AlekhyaReddy: ~/OS  X + v
├── task_resume_job
├── task_resume_job.c
├── task_sigint_forward
├── task_sigint_forward.c
├── task_sigint_forward.c.save
├── task_signal_actions
├── task_signal_actions.c
├── task_signal_behavior
├── task_signal_behavior.c
├── task_sigstop
├── task_sigstop.c
├── task_valgrind
├── task_valgrind.c
├── task_valgrind_leak
├── task_valgrind_leak.c
├── test_report.txt
├── test_script.sh
├── tests
│   ├── test_executor.c
│   └── test_parser.c
├── valgrind_report.txt
├── week4_T2
├── week4_T2.c
├── week_5_double_quotes.c
└── week_5double_quotes

9 directories, 100 files
alekhya@AlekhyaReddy:~/OSSP$ nano README.md
alekhya@AlekhyaReddy:~/OSSP$ nano Makefile
alekhya@AlekhyaReddy:~/OSSP$ make
mkdir -p obj
gcc -Wall -Wextra -Iinclude -c src/main.c -o obj/main.o
mkdir -p obj
gcc -Wall -Wextra -Iinclude -c src/shell.c -o obj/shell.o
mkdir -p obj
gcc -Wall -Wextra -Iinclude -c src/parser.c -o obj/parser.o
mkdir -p obj
gcc -Wall -Wextra -Iinclude -c src/executor.c -o obj/executor.o
mkdir -p obj
gcc -Wall -Wextra -Iinclude -c src/builtin.c -o obj/builtin.o
mkdir -p bin
gcc obj/main.o obj/shell.o obj/parser.o obj/executor.o obj/builtin.o -o bin/my_shell
/usr/bin/ld: /usr/lib/gcc/x86_64-linux-gnu/13/../../../../x86_64-linux-gnu/Scrt1.o: in function `_start':
(.text+0x1b): undefined reference to `main'
collect2: error: ld returned 1 exit status
make: *** [Makefile:18: bin/my_shell] Error 1
alekhya@AlekhyaReddy:~/OSSP$ make clean

```



alekhya@AlekhyaReddy: ~/OS



make: *** [Makefile:18: bin/my_shell] Error 1

alekhya@AlekhyaReddy:~/OSSP\$ make clean

rm -rf obj/*.o bin/my_shell

alekhya@AlekhyaReddy:~/OSSP\$ find src include tests scripts -type f

src/parser.c

src/executor.c

src/builtin.c

src/main.c

src/fork_exec.c

src/shell.c

include/shell.h

include/builtin.h

include/executor.h

include/parser.h

tests/test_parser.c

tests/test_executor.c

scripts/run.sh

alekhya@AlekhyaReddy:~/OSSP\$ git status

On branch master

No commits yet

Untracked files:

(use "git add <file>..." to include in what will be committed)

.gitignore

LICENSE

Makefile

README.md

assets/

bin/

calculator.c

calculator.h

calculator.o

calculator_test

calculator_test.c

client

client.c

docs/

error.log

expected_report.txt

include/

ls_grep

ls_grep.c

main_loop

main_loop.c

main_module.c

main_module.o



alekhya@Alekhyaireddy: ~/OS



```
main_module.o
module_test
producer_consumer
producer_consumer.c
scripts/
server
server.c
signal_handler
signal_handler.c
src/
task1_wait
task1_wait.c
task1_wait.c.save
task2_quotes
task2_quotes.c
task2_single_quotes
task2_single_quotes.c
task3_double_quotes
task3_double_quotes.c
task3_fork
task3_fork.c
task3_input
task3_input.c
task4_escape
task4_escape.c
task4_history
task4_history.c
task4_states
task4_states.c
task5_memory
task5_memory.c
task5_process
task5_process.c
task_error_handling
task_error_handling.c
task_large_pipeline
task_large_pipeline.c
task_process_groups
task_process_groups.c
task_resume_job
task_resume_job.c
task_sigint_forward
task_sigint_forward.c
task_sigint_forward.c.save
task_signal_actions
task_signal_actions.c
task_signal_behavior
```

alekhya@Alekhyareddy: ~/OS

```
task_signal_behavior.c
task_sigstsp
task_sigstsp.c
task_valgrind
task_valgrind.c
task_valgrind_leak
task_valgrind_leak.c
test_report.txt
test_script.sh
tests/
valgrind_report.txt
week4_T2
week4_T2.c
week_5_double_quotes.c
week_5double_quotes
```

nothing added to commit but untracked files present (use "git add" to track)

```
alekhya@Alekhyareddy:~/OSSP$ git add .
```

```
alekhya@Alekhyareddy:~/OSSP$ git commit -m "Organize project and validate deliverables"
```

```
[master (root-commit) 8a79dd1] Organize project and validate deliverables
```

```
Committer: ALEKHYAREDDY <alekhya@Alekhyareddy.localdomain>
```

Your name and email address were configured automatically based on your username and hostname. Please check that they are accurate. You can suppress this message by setting them explicitly. Run the following command and follow the instructions in your editor to edit your configuration file:

```
git config --global --edit
```

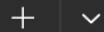
After doing this, you may fix the identity used for this commit with:

```
git commit --amend --reset-author
```

```
98 files changed, 3242 insertions(+)
create mode 100644 .gitignore
create mode 100644 LICENSE
create mode 100644 Makefile
create mode 100644 README.md
create mode 100644 assets/architecture.png
create mode 100755 bin/fork_exec
create mode 100644 calculator.c
create mode 100644 calculator.h
create mode 100644 calculator.o
create mode 100755 calculator_test
create mode 100644 calculator_test.c
create mode 100755 client
create mode 100644 client.c
```



alekhya@Alekhyaireddy: ~/OS



```
create mode 100644 docs/syscalls.txt
create mode 100644 error.log
create mode 100644 expected_report.txt
create mode 100644 include/builtin.h
create mode 100644 include/executor.h
create mode 100644 include/parser.h
create mode 100644 include/shell.h
create mode 100755 ls_grep
create mode 100644 ls_grep.c
create mode 100755 main_loop
create mode 100644 main_loop.c
create mode 100644 main_module.c
create mode 100644 main_module.o
create mode 100755 module_test
create mode 100755 producer_consumer
create mode 100644 producer_consumer.c
create mode 100644 scripts/run.sh
create mode 100755 server
create mode 100644 server.c
create mode 100755 signal_handler
create mode 100644 signal_handler.c
create mode 100644 src/builtin.c
create mode 100644 src/executor.c
create mode 100644 src/fork_exec.c
create mode 100644 src/main.c
create mode 100644 src/parser.c
create mode 100644 src/shell.c
create mode 100755 task1_wait
create mode 100644 task1_wait.c
create mode 100644 task1_wait.c.save
create mode 100755 task2_quotes
create mode 100644 task2_quotes.c
create mode 100755 task2_single_quotes
create mode 100644 task2_single_quotes.c
create mode 100755 task3_double_quotes
create mode 100644 task3_double_quotes.c
create mode 100755 task3_fork
create mode 100644 task3_fork.c
create mode 100755 task3_input
create mode 100644 task3_input.c
create mode 100755 task4_escape
create mode 100644 task4_escape.c
create mode 100755 task4_history
create mode 100644 task4_history.c
create mode 100755 task4_states
create mode 100644 task4_states.c
create mode 100755 task5_memory
```

alekhya@Alekhyaireddy: ~/OS

```
create mode 100755 week_5double_quotes
alekhya@Alekhyaireddy:~/OSSP$ git tag -a v1.0.0 -m "Initial project release"
alekhya@Alekhyaireddy:~/OSSP$ git tag
v1.0.0
alekhya@Alekhyaireddy:~/OSSP$ git show v1.0.0
tag v1.0.0
Tagger: ALEKHYAREDDY <alekhya@Alekhyaireddy.localdomain>
Date: Tue Sep 22 14:46:25 2026 +0000

Initial project release

commit 8a79dd12c85d0b1d2d344bcb0653ab06a40934c2 (HEAD -> master, tag: v1.0.0)
Author: ALEKHYAREDDY <alekhya@Alekhyaireddy.localdomain>
Date: Tue Sep 22 14:46:15 2026 +0000
```

Organize project and validate deliverables

```
diff --git a/.gitignore b/.gitignore
new file mode 100644
index 0000000..e69de29
diff --git a/LICENSE b/LICENSE
new file mode 100644
index 0000000..e69de29
diff --git a/Makefile b/Makefile
new file mode 100644
index 0000000..8de3c36
--- /dev/null
+++ b/Makefile
@@ -0,0 +1,31 @@
+CC = gcc
+CFLAGS = -Wall -Wextra -Iinclude
+
+TARGET = bin/my_shell
+
+SOURCES = src/main.c \
+         src/shell.c \
+         src/parser.c \
+         src/executor.c \
+         src/builtin.c
+
+OBJECTS = $(SOURCES:src/%.c=obj/%.o)
+
+all: $(TARGET)
+
+$(TARGET): $(OBJECTS)
+    mkdir -p bin
+    $(CC) $(OBJECTS) -o $(TARGET)
```



alekhya@Alekhyareddy: ~/OS



Tagger: ALEKHYAREDDY <alekhya@Alekhyareddy.localdomain>

Date: Tue Sep 22 14:46:25 2026 +0000

Initial project release

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Author: ALEKHYAREDDY <alekhya@Alekhyareddy.localdomain>

Date: Tue Sep 22 14:46:15 2026 +0000

Organize project and validate deliverables

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new file mode 100644

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--- /dev/null

+++ b/Makefile

@@ -0,0 +1,31 @@

+CC = gcc

+CFLAGS = -Wall -Wextra -Iinclude

+

+TARGET = bin/my_shell

+

+SOURCES = src/main.c \

+ src/shell.c \

+ src/parser.c \

+ src/executor.c \

+ src/builtin.c

+

+OBJECTS = \$(SOURCES:src/%.c=obj/%.o)

+

+all: \$(TARGET)

+

+\$\$(TARGET): \$(OBJECTS)

+ mkdir -p bin

+ \$(CC) \$(OBJECTS) -o \$(TARGET)

+

+obj/%.o: src/%.c

+ mkdir -p obj

+ \$(CC) \$(CFLAGS) -c \$< -o \$@

+

.

commit 8a79dd12c85d0b1d2d344bcb0653ab06a40934c2 (HEAD -> master, tag: v1.0.0)

Author: ALEKHYAREDDY <alekhya@Alekhyaireddy.localdomain>

Date: Tue Sep 22 14:46:15 2026 +0000

Organize project and validate deliverables

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diff --git a/Makefile b/Makefile

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--- /dev/null

+++ b/Makefile

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+

+TARGET = bin/my_shell

+

+SOURCES = src/main.c \

+ src/shell.c \

+ src/parser.c \

+ src/executor.c \

+ src/builtin.c

+

+OBJECTS = \$(SOURCES:src/%.c=obj/%.o)

+

+all: \$(TARGET)

+

+\$\$(TARGET): \$(OBJECTS)

+ mkdir -p bin

+ \$(CC) \$(OBJECTS) -o \$(TARGET)

+

+obj/%.o: src/%.c

+ mkdir -p obj

+ \$(CC) \$(CFLAGS) -c \$< -o \$@

+

:

Tagger: ALEKHYAREDDY <alekhya@Alekhyaireddy.localdomain>

Date: Tue Sep 22 14:46:25 2026 +0000

Initial project release

commit 8a79dd12c85d0b1d2d344bcb0653ab06a40934c2 (HEAD -> master, tag: v1.0.0)

Author: ALEKHYAREDDY <alekhya@Alekhyareddy.localdomain>

Date: Tue Sep 22 14:46:15 2026 +0000

Organize project and validate deliverables

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--- /dev/null

+++ b/Makefile

@@ -0,0 +1,31 @@

+CC = gcc

+CFLAGS = -Wall -Wextra -Iinclude

+

+TARGET = bin/my_shell

+

+SOURCES = src/main.c \

+ src/shell.c \

+ src/parser.c \

+ src/executor.c \

+ src/builtin.c

+

+OBJECTS = \$(SOURCES:src/%.c=obj/%.o)

+

+all: \$(TARGET)

+

+\$\$(TARGET): \$(OBJECTS)

+ mkdir -p bin

+ \$(CC) \$(OBJECTS) -o \$(TARGET)

+

+obj/%.o: src/%.c

+ mkdir -p obj

+ \$(CC) \$(CFLAGS) -c \$< -o \$@

+

:

Tagger: ALEKHYAREDDY <alekhya@Alekhyareddy.localdomain>

tag v1.0.0

Tagger: ALEKHYAREDDY <alekhya@Alekhyareddy.localdomain>

Date: Tue Sep 22 14:46:25 2026 +0000

```
commit 8a79dd12c85d0b1d2d344bcb0653ab06a40934c2 (HEAD -> master, tag: v1.0.0)
```

```
Author: ALEKHYAREDDY <alekhya@Alekhyareddy.localdomain>
```

```
Date: Tue Sep 22 14:46:15 2026 +0000
```

Organize project and validate deliverables

```
diff --git a/.gitignore b/.gitignore
```

```
new file mode 100644
```

```
index 0000000..e69de29
```

```
diff --git a/LICENSE b/LICENSE
```

```
new file mode 100644
```

```
index 0000000..e69de29
```

```
diff --git a/Makefile b/Makefile
```

```
new file mode 100644
```

```
index 0000000..8de3c36
```

```
--- /dev/null
```

```
+++ b/Makefile
```

```
@@ -0,0 +1,31 @@
```

```
+CC = gcc
```

```
+CFLAGS = -Wall -Wextra -Iinclude
```

```
+
```

```
+TARGET = bin/my_shell
```

```
+
```

```
+SOURCES = src/main.c \
```

```
+          src/shell.c \
```

```
+          src/parser.c \
```

```
+          src/executor.c \
```

```
+          src/builtin.c
```

```
+
```

```
+OBJECTS = $(SOURCES:src/%.c=obj/%.o)
```

```
+
```

```
+all: $(TARGET)
```

```
+
```

```
+$$(TARGET): $(OBJECTS)
```

```
+      mkdir -p bin
```

```
+      $(CC) $(OBJECTS) -o $(TARGET)
```

```
+
```

```
+obj/%.o: src/%.c
```

```
+      mkdir -p obj
```

```
+      $(CC) $(CFLAGS) -c $< -o $@
```

```
log file: how v1.0.0
```

```
tag v1.0.0
```

```
Tagger: ALEKHYAREDDY <alekhya@Alekhyareddy.localdomain>
```

```
Date: Tue Sep 22 14:46:25 2026 +0000
```

Initial project release



alekhya@Alekhyaireddy: ~/OS



[1]+ Stopped git show v1.0.0

alekhya@Alekhyaireddy:~/OSSP\$ git status

On branch master

nothing to commit, working tree clean

alekhya@Alekhyaireddy:~/OSSP\$ tree

```
.
├── LICENSE
├── Makefile
├── README.md
├── assets
│   └── architecture.png
├── bin
│   └── fork_exec
├── calculator.c
├── calculator.h
├── calculator.o
├── calculator_test
├── calculator_test.c
├── client
├── client.c
├── client_to_server
├── docs
│   ├── architecture.txt
│   ├── design.md
│   ├── design_decisions.txt
│   ├── execution_flow.txt
│   ├── report.pdf
│   └── syscalls.txt
├── error.log
├── expected_report.txt
├── include
│   ├── builtin.h
│   ├── executor.h
│   ├── parser.h
│   └── shell.h
├── ls_grep
├── ls_grep.c
├── main_loop
├── main_loop.c
├── main_module.c
├── main_module.o
├── module_test
├── obj
├── producer_consumer
├── producer_consumer.c
└── scripts
```



alekhya@Alekhyareddy: ~/OS



scripts

└─ run.sh

server

server.c

server_to_client

signal_handler

signal_handler.c

src

└─ builtin.c

└─ executor.c

└─ fork_exec.c

└─ main.c

└─ parser.c

└─ shell.c

task1_wait

task1_wait.c

task1_wait.c.save

task2_quotes

task2_quotes.c

task2_single_quotes

task2_single_quotes.c

task3_double_quotes

task3_double_quotes.c

task3_fork

task3_fork.c

task3_input

task3_input.c

task4_escape

task4_escape.c

task4_history

task4_history.c

task4_states

task4_states.c

task5_memory

task5_memory.c

task5_process

task5_process.c

task_error_handling

task_error_handling.c

task_large_pipeline

task_large_pipeline.c

task_process_groups

task_process_groups.c

task_resume_job

task_resume_job.c

task_sigint_forward

task_sigint_forward.c

```
task_sigint_forward.c.save
task_signal_actions
task_signal_actions.c
task_signal_behavior
task_signal_behavior.c
task_sigstop
task_sigstop.c
task_valgrind
task_valgrind.c
task_valgrind_leak
task_valgrind_leak.c
test_report.txt
test_script.sh
tests
├── test_executor.c
├── test_parser.c
valgrind_report.txt
week4_T2
week4_T2.c
week_5_double_quotes.c
week_5double_quotes
```

9 directories, 99 files

alekhya@Alekhyareddy:~/OSSP\$ |

4 . To create demo scenarios, validate project features, rehearse the presentation, demonstrate pipelines, demonstrate signal handling, and perform the final project review.

Aim : To verify the complete project functionality through practical demonstrations, test important features such as pipelines and signals, and prepare the project for final presentation and evaluation.

Output :

TERMINAL - 1 :



alekhya@Alekhyareddy: ~/OS



alekhya@Alekhyareddy:~\$ cd ~/OSSP

alekhya@Alekhyareddy:~/OSSP\$ tree

```
.
├── LICENSE
├── Makefile
├── README.md
├── assets
│   └── architecture.png
├── bin
│   └── fork_exec
├── calculator.c
├── calculator.h
├── calculator.o
├── calculator_test
├── calculator_test.c
├── client
├── client.c
├── client_to_server
├── docs
│   ├── architecture.txt
│   ├── design.md
│   ├── design_decisions.txt
│   ├── execution_flow.txt
│   ├── report.pdf
│   └── syscalls.txt
├── error.log
├── expected_report.txt
├── include
│   ├── builtin.h
│   ├── executor.h
│   ├── parser.h
│   └── shell.h
├── ls_grep
├── ls_grep.c
├── main_loop
├── main_loop.c
├── main_module.c
├── main_module.o
├── module_test
├── obj
├── producer_consumer
├── producer_consumer.c
├── scripts
│   └── run.sh
├── server
├── server.c
├── server_to_client
├── signal_handler
├── signal_handler.c
├── src
│   └── builtin.c
```



alekhya@Alekhyareddy: ~/OS



```
├── main.c
├── parser.c
├── shell.c
├── task1_wait
├── task1_wait.c
├── task1_wait.c.save
├── task2_quotes
├── task2_quotes.c
├── task2_single_quotes
├── task2_single_quotes.c
├── task3_double_quotes
├── task3_double_quotes.c
├── task3_fork
├── task3_fork.c
├── task3_input
├── task3_input.c
├── task4_escape
├── task4_escape.c
├── task4_history
├── task4_history.c
├── task4_states
├── task4_states.c
├── task5_memory
├── task5_memory.c
├── task5_process
├── task5_process.c
├── task_error_handling
├── task_error_handling.c
├── task_large_pipeline
├── task_large_pipeline.c
├── task_process_groups
├── task_process_groups.c
├── task_resume_job
├── task_resume_job.c
├── task_sigint_forward
├── task_sigint_forward.c
├── task_sigint_forward.c.save
├── task_signal_actions
├── task_signal_actions.c
├── task_signal_behavior
├── task_signal_behavior.c
├── task_sigstsp
├── task_sigstsp.c
├── task_valgrind
├── task_valgrind.c
├── task_valgrind_leak
├── task_valgrind_leak.c
├── test_report.txt
├── test_script.sh
└── tests
    ├── test_executor.c
```

```

alekhya@Alekhyaireddy: ~/OS
├─ test_parser.c
├─ valgrind_report.txt
├─ week4_T2
├─ week4_T2.c
├─ week_5_double_quotes.c
└─ week_5double_quotes

9 directories, 99 files
alekhya@Alekhyaireddy:~/OSSP$ make clean
rm -rf obj/*.o bin/my_shell
alekhya@Alekhyaireddy:~/OSSP$ make
mkdir -p obj
gcc -Wall -Wextra -Iinclude -c src/main.c -o obj/main.o
mkdir -p obj
gcc -Wall -Wextra -Iinclude -c src/shell.c -o obj/shell.o
mkdir -p obj
gcc -Wall -Wextra -Iinclude -c src/parser.c -o obj/parser.o
mkdir -p obj
gcc -Wall -Wextra -Iinclude -c src/executor.c -o obj/executor.o
mkdir -p obj
gcc -Wall -Wextra -Iinclude -c src/builtin.c -o obj/builtin.o
mkdir -p bin
gcc obj/main.o obj/shell.o obj/parser.o obj/executor.o obj/builtin.o -o bin/my_shell
/usr/bin/ld: /usr/lib/gcc/x86_64-linux-gnu/13/../../../../x86_64-linux-gnu/Scrt1.o: in function '_start':
(.text+0x1b): undefined reference to 'main'
collect2: error: ld returned 1 exit status
make: *** [Makefile:18: bin/my_shell] Error 1
alekhya@Alekhyaireddy:~/OSSP$ ./bin/my_shell
-bash: ./bin/my_shell: No such file or directory
alekhya@Alekhyaireddy:~/OSSP$ ls
LICENSE
Makefile
README.md
assets
bin
calculator.c
calculator.h
calculator.o
calculator_test
calculator_test.c
client
client.c
client_to_server
docs
error.log
expected_report.txt
include
ls_grep
alekhya@Alekhyaireddy:~/OSSP$ pwd
/home/alekhya/OSSP
alekhya@Alekhyaireddy:~/OSSP$ whoami
alekhya
ls_grep.c
main_loop
main_loop.c
main_module.c
main_module.o
module_test
obj
producer_consumer
producer_consumer.c
scripts
server
server.c
server_to_client
signal_handler
signal_handler.c
src
task1_wait
task1_wait.c
task1_wait.c.save
task2_quotes
task2_quotes.c
task2_single_quotes
task2_single_quotes.c
task3_double_quotes
task3_double_quotes.c
task3_fork
task3_fork.c
task3_input
task3_input.c
task4_escape
task4_escape.c
task4_history
task4_history.c
task4_states
task4_states.c
task5_memory
task5_memory.c
task5_process
task5_process.c
task_error_handling
task_error_handling.c
task_large_pipeline
task_large_pipeline.c
task_process_groups
task_process_groups.c
task_resume_job
task_resume_job.c
task_sigint_forward
task_sigint_forward.c
task_sigint_forward.c.save
task_signal_actions
task_signal_actions.c
task_signal_behavior
task_signal_behavior.c
task_sigsttp
task_sigsttp.c
task_valgrind
task_valgrind.c
task_valgrind_leak
task_valgrind_leak.c
test_report.txt
test_script.sh
tests
valgrind_report.txt
week4_T2
week4_T2.c
week_5_double_quotes.c
week_5double_quotes

```

```

alekhya@Alekhyaireddy: ~/OS
alekhya@Alekhyaireddy:~/OSSP$ whoami
alekhya
alekhya@Alekhyaireddy:~/OSSP$ ls -l | grep ".c"
-rw-r--r-- 1 alekhya alekhya 159 Sep 22 13:41 calculator.c
-rw-r--r-- 1 alekhya alekhya 130 Sep 22 13:39 calculator.h
-rw-r--r-- 1 alekhya alekhya 1464 Sep 22 13:44 calculator.o
-rwxr-xr-x 1 alekhya alekhya 16064 Sep 22 14:14 calculator_test
-rw-r--r-- 1 alekhya alekhya 355 Sep 22 14:14 calculator_test.c
-rwxr-xr-x 1 alekhya alekhya 16392 Sep 18 03:51 client
-rw-r--r-- 1 alekhya alekhya 1018 Sep 18 03:50 client.c
-rw-r--r-- 1 alekhya alekhya 90 Sep 17 06:52 client_to_server
-rw-r--r-- 1 alekhya alekhya 4096 Sep 22 14:37 docs
-rw-r--r-- 1 alekhya alekhya 777 Sep 22 14:17 expected_report.txt
-rw-r--r-- 1 alekhya alekhya 465 Sep 22 13:42 include
-rw-r--r-- 1 alekhya alekhya 1376 Sep 18 03:32 ls_grep.c
-rw-r--r-- 1 alekhya alekhya 420 Aug 13 11:45 main_loop.c
-rw-r--r-- 1 alekhya alekhya 726 Sep 18 03:58 main_module.c
-rwxr-xr-x 1 alekhya alekhya 16488 Sep 18 03:26 producer_consumer
-rw-r--r-- 1 alekhya alekhya 1677 Sep 18 03:26 producer_consumer.c
-rw-r--r-- 1 alekhya alekhya 1275 Sep 18 03:48 server.c
-rw-r--r-- 1 alekhya alekhya 0 Sep 18 03:52 server_to_client
-rw-r--r-- 1 alekhya alekhya 726 Sep 18 03:58 signal_handler.c
-rwxr-xr-x 1 alekhya alekhya 4096 Aug 4 05:32 src
-rw-r--r-- 1 alekhya alekhya 1851 Aug 17 13:42 task1_wait.c
-rw-r--r-- 1 alekhya alekhya 1851 Aug 17 13:46 task1_wait.c.save
-rw-r--r-- 1 alekhya alekhya 2380 Aug 13 11:56 task2_quotes.c
-rw-r--r-- 1 alekhya alekhya 1705 Sep 18 04:13 task2_single_quotes.c
-rw-r--r-- 1 alekhya alekhya 3056 Sep 18 04:38 task3_double_quotes.c
-rw-r--r-- 1 alekhya alekhya 943 Aug 17 08:09 task3_fork.c
-rw-r--r-- 1 alekhya alekhya 1661 Aug 7 08:09 task3_input.c
-rwxr-xr-x 1 alekhya alekhya 16488 Aug 17 04:59 task4_escape
-rw-r--r-- 1 alekhya alekhya 227 Aug 17 08:56 task4_escape.c
-rw-r--r-- 1 alekhya alekhya 3976 Aug 7 08:19 task4_history.c
-rw-r--r-- 1 alekhya alekhya 954 Aug 17 13:32 task4_states.c
-rw-r--r-- 1 alekhya alekhya 2612 Aug 17 08:56 task5_memory.c
-rwxr-xr-x 1 alekhya alekhya 16696 Aug 17 05:03 task5_process
-rw-r--r-- 1 alekhya alekhya 2092 Aug 17 05:03 task5_process.c
-rw-r--r-- 1 alekhya alekhya 1501 Sep 22 13:50 task_error_handling.c
-rw-r--r-- 1 alekhya alekhya 2705 Sep 22 14:27 task_large_pipeline.c
-rwxr-xr-x 1 alekhya alekhya 16648 Sep 21 05:39 task_process_groups
-rw-r--r-- 1 alekhya alekhya 2450 Sep 21 05:39 task_process_groups.c
-rw-r--r-- 1 alekhya alekhya 1959 Sep 21 05:15 task_resume_job.c
-rw-r--r-- 1 alekhya alekhya 2131 Sep 21 05:09 task_sigint_forward.c
-rwxr-xr-x 1 alekhya alekhya 16368 Sep 21 04:56 task_sigint_forward.c.save
-rw-r--r-- 1 alekhya alekhya 1550 Sep 21 04:56 task_signal_actions.c
-rw-r--r-- 1 alekhya alekhya 1617 Sep 21 05:54 task_signal_behavior.c
-rw-r--r-- 1 alekhya alekhya 2602 Sep 21 05:29 task_sigsttp.c
-rw-r--r-- 1 alekhya alekhya 769 Sep 22 14:03 task_valgrind.c
-rw-r--r-- 1 alekhya alekhya 443 Sep 22 14:07 task_valgrind_leak.c

```

```

-rw-r--r-- 1 alekhya alekhya 662 Aug 17 13:53 week4_T2.c
-rw-r--r-- 1 alekhya alekhya 3056 Sep 18 04:11 week_5_double_quotes.c
alekhya@AlekhyaReddy:~/OSSP$ ps
  PID TTY          TIME CMD
  508 pts/0        00:00:00 bash
  558 pts/0        00:00:00 ps
alekhya@AlekhyaReddy:~/OSSP$ ps -o pid,ppid,pgid,state,cmd
  PID   PPID    PGID S  CMD
   508    502    508 S  -bash
   561    508    561 R  ps -o pid,ppid,pgid,state,cmd
alekhya@AlekhyaReddy:~/OSSP$ gcc -Wall -Wextra task_signal_behavior.c -o task_signal_behavior
alekhya@AlekhyaReddy:~/OSSP$ ./task_signal_behavior
Signal Behavior Analysis
-----
Process PID       : 569
Process Group ID  : 569
Parent PID        : 508

Signal handlers registered.
Blocking SIGINT for 10 seconds...
SIGINT is now blocked.
Press Ctrl+C during these 10 seconds.

Unblocking SIGINT...
SIGINT is now unblocked.

Waiting for signals...
Use another terminal to send SIGUSR1 or SIGTERM.

SIGUSR1 received: User-defined signal delivered.

SIGTERM received: Termination signal delivered.

```

TERMINAL - 2 :

```

alekhya@AlekhyaReddy: ~
alekhya@AlekhyaReddy:~$ ps -ef | grep task_signal_behavior
alekhya      569      508  0 14:56 pts/0        00:00:00 ./task_signal_behavior
alekhya      591      574  0 14:56 pts/2        00:00:00 grep --color=auto task_signal_behavior
alekhya@AlekhyaReddy:~$ kill -SIGUSR1 569
alekhya@AlekhyaReddy:~$ kill -SIGTERM 569
alekhya@AlekhyaReddy:~$ ./task_resume_job
-bash: ./task_resume_job: No such file or directory
alekhya@AlekhyaReddy:~$ jobs
alekhya@AlekhyaReddy:~$ fg
-bash: fg: current: no such job
alekhya@AlekhyaReddy:~$ make test
make: *** No rule to make target 'test'.  Stop.
alekhya@AlekhyaReddy:~$ valgrind --leak-check=full --show-leak-kinds=all ./bin/my_shell
valgrind: ./bin/my_shell: No such file or directory
alekhya@AlekhyaReddy:~$ git status
fatal: not a git repository (or any of the parent directories): .git
alekhya@AlekhyaReddy:~$ git log --oneline -5
fatal: not a git repository (or any of the parent directories): .git
alekhya@AlekhyaReddy:~$ git tag
fatal: not a git repository (or any of the parent directories): .git
alekhya@AlekhyaReddy:~$ git show v1.0.0
fatal: not a git repository (or any of the parent directories): .git
alekhya@AlekhyaReddy:~$ tree
.
├── File1.txt
├── File2.txt
├── LICENSE
├── Makefile
├── OSLab
│   ├── hello
│   └── newfile.txt
├── OSSP
│   ├── LICENSE
│   ├── Makefile
│   ├── README.md
│   ├── assets
│   │   └── architecture.png
│   ├── bin
│   │   └── fork_exec
│   ├── calculator.c
│   ├── calculator.h
│   ├── calculator.o
│   └── calculator_test

```



alekhya@Alekhyareddy: ~



- calculator_test
- calculator_test.c
- client
- client.c
- client_to_server
- docs
 - architecture.txt
 - design.md
 - design_decisions.txt
 - execution_flow.txt
 - report.pdf
 - syscalls.txt
- error.log
- expected_report.txt
- include
 - builtin.h
 - executor.h
 - parser.h
 - shell.h
- ls_grep
- ls_grep.c
- main_loop
- main_loop.c
- main_module.c
- main_module.o
- module_test
- obj
 - builtin.o
 - executor.o
 - main.o
 - parser.o
 - shell.o
- producer_consumer
- producer_consumer.c
- scripts
 - run.sh
- server
- server.c
- server_to_client
- signal_handler
- signal_handler.c
- src
 - builtin.c



alekhya@Alekhyareddy: ~



```
| parser.c
| shell.c
```

```
task1_wait
```

```
task1_wait.c
```

```
task1_wait.c.save
```

```
task2_quotes
```

```
task2_quotes.c
```

```
task2_single_quotes
```

```
task2_single_quotes.c
```

```
task3_double_quotes
```

```
task3_double_quotes.c
```

```
task3_fork
```

```
task3_fork.c
```

```
task3_input
```

```
task3_input.c
```

```
task4_escape
```

```
task4_escape.c
```

```
task4_history
```

```
task4_history.c
```

```
task4_states
```

```
task4_states.c
```

```
task5_memory
```

```
task5_memory.c
```

```
task5_process
```

```
task5_process.c
```

```
task_error_handling
```

```
task_error_handling.c
```

```
task_large_pipeline
```

```
task_large_pipeline.c
```

```
task_process_groups
```

```
task_process_groups.c
```

```
task_resume_job
```

```
task_resume_job.c
```

```
task_sigint_forward
```

```
task_sigint_forward.c
```

```
task_sigint_forward.c.save
```

```
task_signal_actions
```

```
task_signal_actions.c
```

```
task_signal_behavior
```

```
task_signal_behavior.c
```

```
task_sigtstp
```

```
task_sigtstp.c
```

```
task_valgrind
```



28°C



Search



alekhya@Alekhyareddy: ~



- └─ **command_executor**
 - command_executor.c

- assets**

- └─ **architecture.png**

- bin**

- builtin_practical**

- └─ **builtin_shell**
 - builtin_shell.c

- command_executor**

- command_executor.c

- cow_practical**

- └─ **cow_demo**
 - cow_demo.c
 - cow_demo.cnano

- docs**

- └─ design.md
 - report.pdf

- filecopy**

- filecopy.c

- foreground_job**

- └─ **foreground_job**
 - foreground_job.c

- fork_exec**

- fork_exec.c

- fork_program**

- hello**

- hello.c

- include**

- └─ builtin.h
 - executor.h
 - parser.h
 - shell.h

- job_practical**

- └─ **jobs**
 - jobs.c

- main.c

- main_loop**

- main_loop.c

- memory**

- memory.c

- memory_maps.c.save

- memory_valgrind**

- └─ **memory_demo**

alekhya@Alekhyareddy: ~

```
├── path_resolution.c
├── process
├── process.c
├── scripts
│   └── run.sh
├── src
│   ├── builtin.c
│   ├── executor.c
│   ├── main.c
│   ├── parser.c
│   └── shell.c
├── task1
├── task1.c
├── tests
│   ├── test_executor.c
│   └── test_parser.c
├── variable_expansion
│   ├── variable_expand
│   └── variable_expand.c
├── waitpid_practical
│   ├── waitpid_demo
│   └── waitpid_demo.c
├── week6
│   ├── escape
│   ├── escape.c
│   ├── my file.txt
│   ├── parser_test.sh
│   └── student file.txt
├── week6_process
│   ├── process
│   └── process.c
```

30 directories, 174 files

alekhya@Alekhyareddy:~\$ git status

fatal: not a git repository (or any of the parent directories): .git

alekhya@Alekhyareddy:~\$ make clean

make: *** No rule to make target 'clean'. Stop.

alekhya@Alekhyareddy:~\$ make

make: *** No targets. Stop.

alekhya@Alekhyareddy:~\$ make test

make: *** No rule to make target 'test'. Stop.

alekhya@Alekhyareddy:~\$./bin/my_shell

-bash: ./bin/my_shell: No such file or directory

```

alekhya@AlekhyaReddy: ~
make: *** No targets. Stop.
alekhya@AlekhyaReddy:~$ make test
make: *** No rule to make target 'test'. Stop.
alekhya@AlekhyaReddy:~$ ./bin/my_shell
-bash: ./bin/my_shell: No such file or directory
alekhya@AlekhyaReddy:~$ ls
File1.txt  Week2          docs           hello          memory         process        variable_expansion
File2.txt  assets         filecopy      hello.c        memory.c       process.c      waitpid_practical
LICENSE    bin           filecopy.c    include        memory_maps.c.save scripts        week6
Makefile   builtin_practical foreground_job job_practical  memory_valgrind src            week6_process
OSLab      command_executor fork_exec     main.c         nano.9019.save task1
OSSP       command_executor.c fork_exec.c   main_loop     obj            task1.c
README.md  cow_practical  fork_program  main_loop.c   path_practical tests
alekhya@AlekhyaReddy:~$ pwd
/home/alekhya
alekhya@AlekhyaReddy:~$ ls -l | grep ".c"
drwxr-xr-x  2 alekhya alekhya  4096 Sep 18 14:25 builtin_practical
-rwxr-xr-x  1 alekhya alekhya 16328 Aug  3 15:16 command_executor
-rw-r--r--  1 alekhya alekhya   847 Aug  3 15:16 command_executor.c
drwxr-xr-x  2 alekhya alekhya  4096 Sep 18 14:53 cow_practical
drwxr-xr-x  2 alekhya alekhya  4096 Aug  1 12:27 docs
-rwxr-xr-x  1 alekhya alekhya 16184 Aug  5 06:58 filecopy
-rw-r--r--  1 alekhya alekhya   734 Aug  5 06:57 filecopy.c
-rwxr-xr-x  1 alekhya alekhya 16320 Aug  3 15:24 fork_exec
-rw-r--r--  1 alekhya alekhya   800 Aug  1 08:54 fork_exec.c
-rw-r--r--  1 alekhya alekhya    78 Aug  3 05:29 hello.c
drwxr-xr-x  2 alekhya alekhya  4096 Aug  1 12:27 include
drwxr-xr-x  2 alekhya alekhya  4096 Sep 19 08:45 job_practical
-rw-r--r--  1 alekhya alekhya   443 Aug  1 12:34 main.c
-rw-r--r--  1 alekhya alekhya   418 Aug  5 06:54 main_loop.c
-rw-r--r--  1 alekhya alekhya   849 Sep 18 04:54 memory.c
-rw-----  1 alekhya alekhya   669 Sep 18 05:15 memory_maps.c.save
drwxr-xr-x  2 alekhya alekhya  4096 Sep 18 14:03 path_practical
-rwxr-xr-x  1 alekhya alekhya 16320 Aug  3 15:02 process
-rw-r--r--  1 alekhya alekhya   833 Aug  3 15:02 process.c
drwxr-xr-x  2 alekhya alekhya  4096 Aug  1 12:28 scripts
drwxr-xr-x  2 alekhya alekhya  4096 Aug  1 12:27 src
-rw-r--r--  1 alekhya alekhya  1002 Aug  4 05:41 task1.c
drwxr-xr-x  2 alekhya alekhya  4096 Sep 18 13:56 waitpid_practical
drwxr-xr-x  2 alekhya alekhya  4096 Sep 18 13:48 week6_process
alekhya@AlekhyaReddy:~$ |

```